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Coarse-grained structure of natural language, or spot the bot

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Our approach

Semantic space of natural language is represented by a set of vectors of n-grams from (literary) texts in the language.

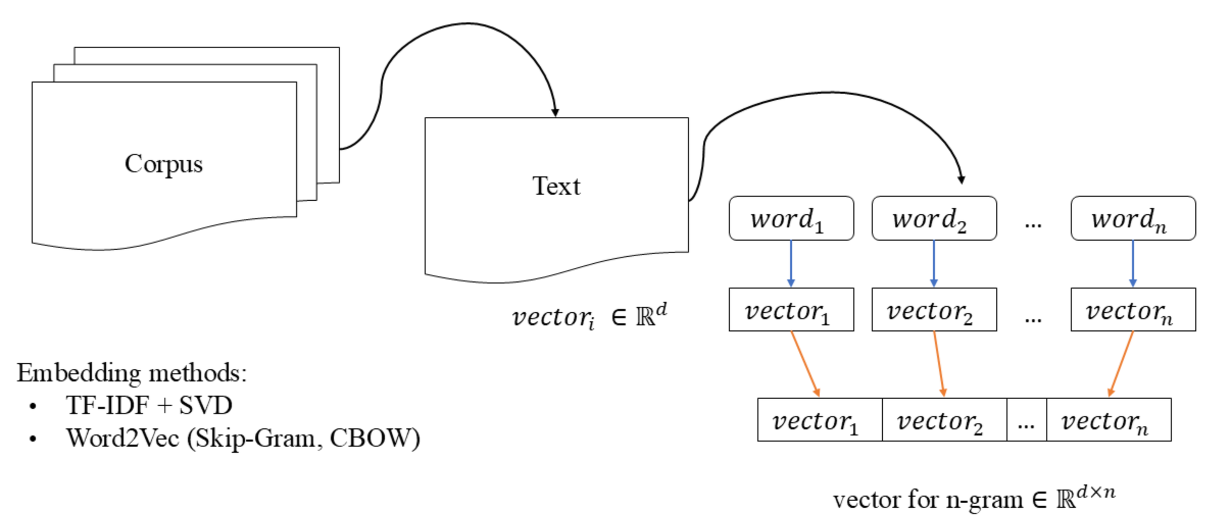

Рис. 1: Method of creating dataset

To analyse the language we employ:

- Clustering - to identify coarse-grained structure of semantic space
- Topological Data Analysis - to find «holes» of the language
- Network Science - to build and analyse networks of concepts

Main Findings

We constructed coarse-grained structure of semantic space using the Wishart algorithm. For bigram clusters usually form with (usually 1) «main» words - for cluster C_i all bigrams $u_j \in C_i$ look like (w_i, w_j) or (w_j, w_i), where w_i is the «main» word of the cluster. For, example, «ночь», «жизнь», «бог» are «main» words.

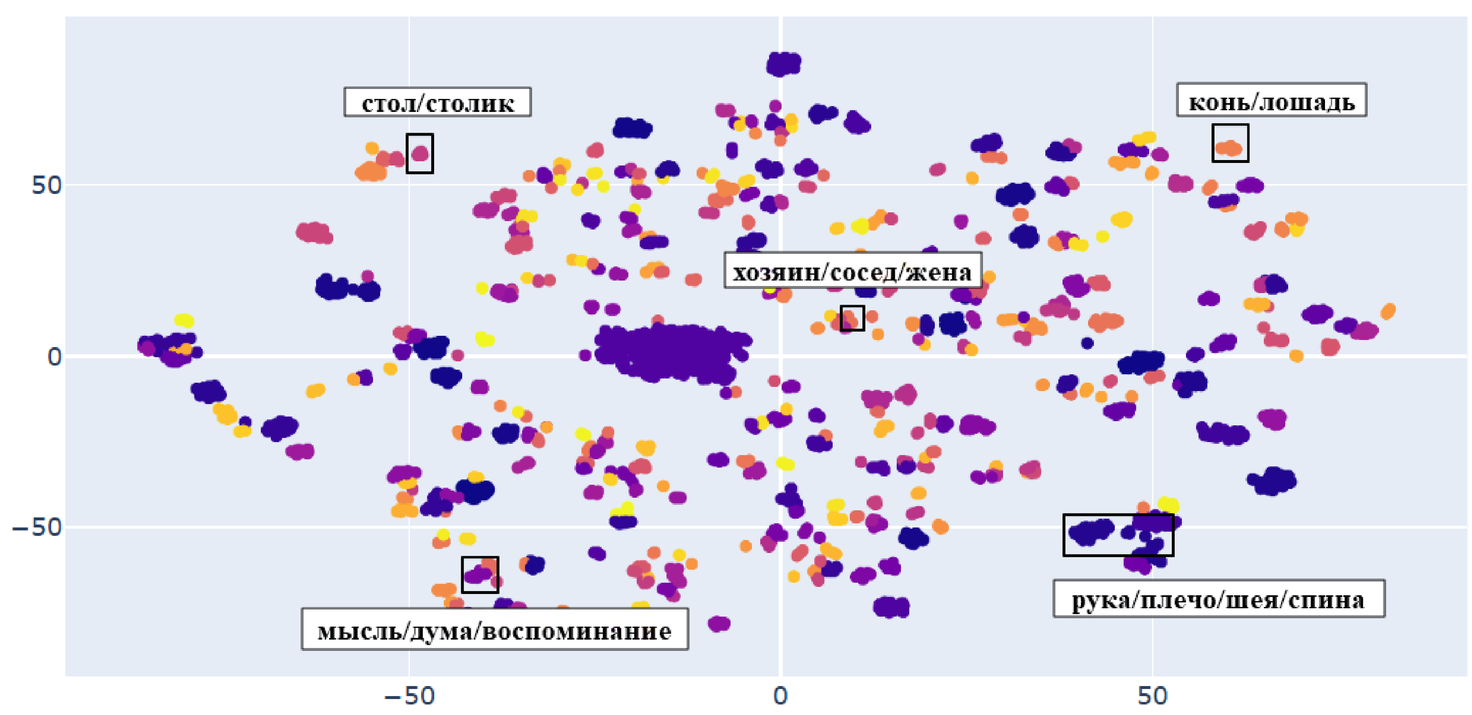

Рис. 2: t-SNE visualization of coarse-grained structure, Russian, CBOW, n=2. Highlighted words - «main» words of the cluster.

«Holes» exist in the language and clusters are not located in them.

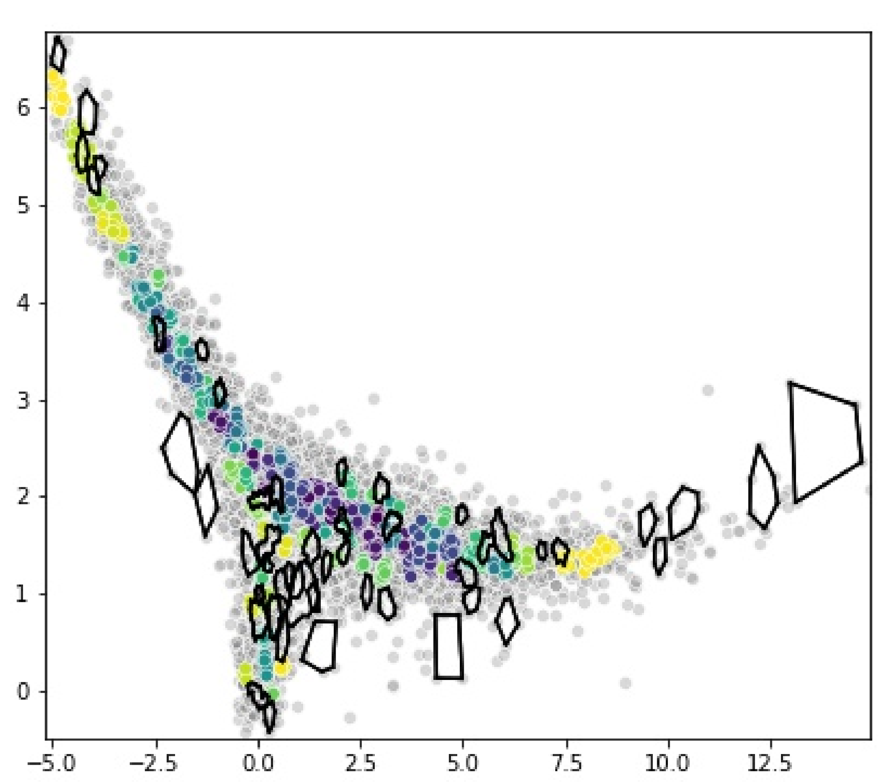

Рис. 3: Coarse-grained structure and «holes», Russian, CBOW, n=1.

Bots vs. Humans

There are statistically significant differences in the coarse-grained structure of semantic spaces of texts written by humans and generated by bots[2].

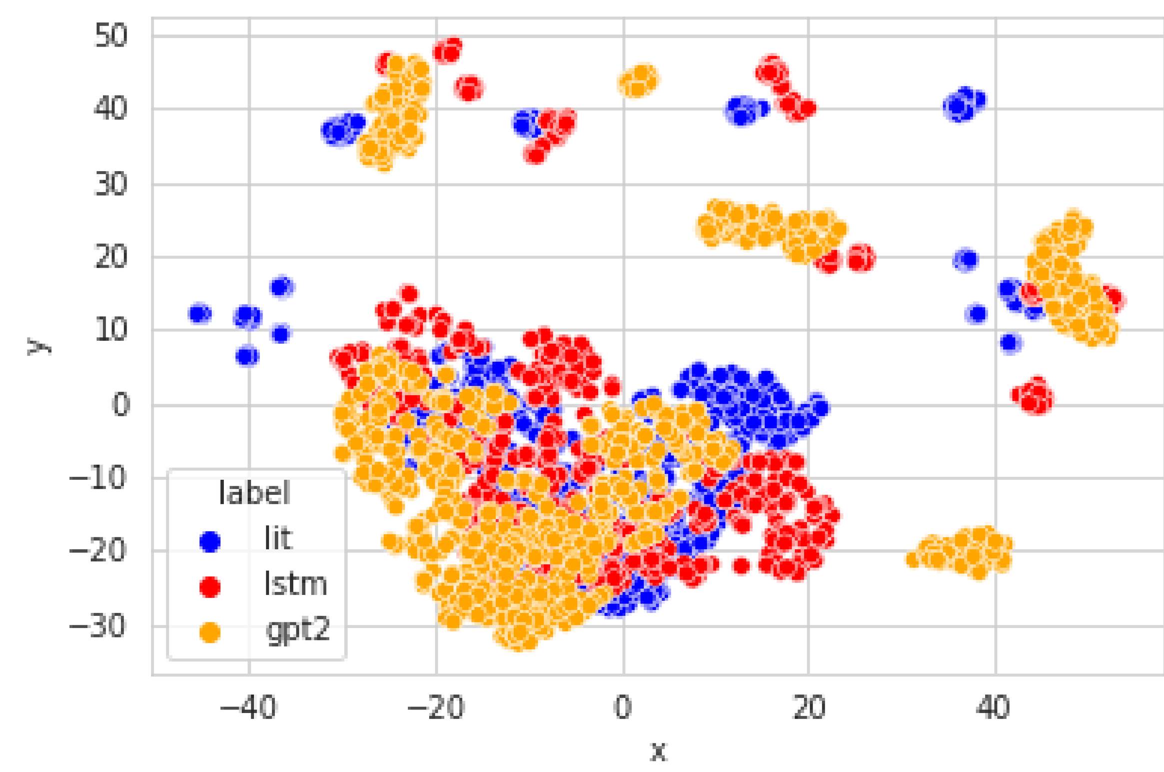

Рис. 4: t-SNE visualization of clusters for bots and humans, Russian, SVD embeddings, n=2

We propose a novel problem statement of bot detection task: to construct training and test sets for bot-generated texts, one randomly samples the set of bots themselves into training and test subsets. So, the model is trained on some bots and tested on others[1].

Employing clustering and entropy-complexity we build classifiers for detecting different bots in 5 languages. For example, the combination of all features gives $F1 = 0.98$ for English language[1].

Таблица 1: F1-score values for unified classification model

	Russian	English	German	French	Vietnamese
SVM	0.82	0.98	0.63	0.82	0.74
DT	0.76	0.85	0.55	0.59	0.62
RF	0.86	0.86	0.57	0.68	0.66

Further research

- Compare «holes» and clusters of diffrent languages
- Build theoretical model based on coarse-grained structure
- Construct network for all the texts in the language
- Compare networks of different languages, employ other methods of building networks

References

[1] Vasilii A. Gromov, Quynh Nhu Dang, Alexandra S. Kogan, Assel Yerbolova Spot the Bot: the Inverse Problems of NLP // PeerJ Computer Science. 2024. Vol. 10.

[2] Gromov V., Kogan A. Spot the Bot: Coarse-Grained Partition of Semantic Paths for Bots and Humans, in Pattern Recognition and Machine Intelligenc, Springer, 2023. P. 348–355.